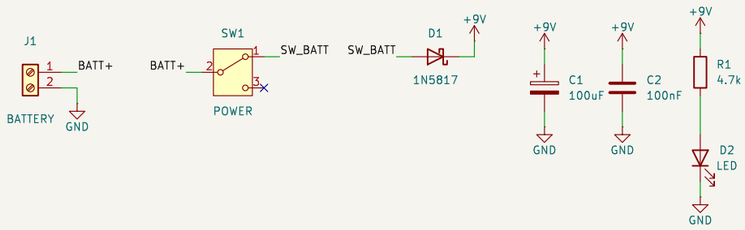
1. Power

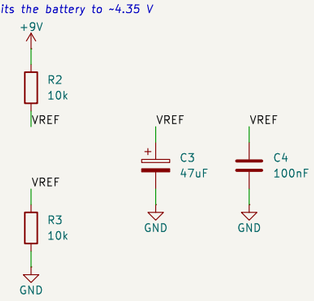
9 V battery -> switch -> reverse-battery diode -> +9V rail



Battery power only.
Never connect to a wall adapter.

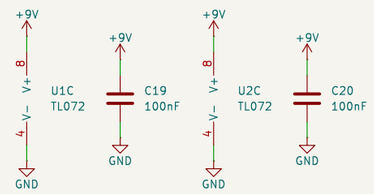
2. Virtual ground (VREF)

Splits the battery to ~4.35 V

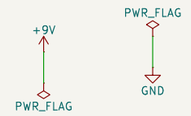


8. Op-amp power + decoupling

C19/C20 sit right next to U1/U2 pin 8



Power flags



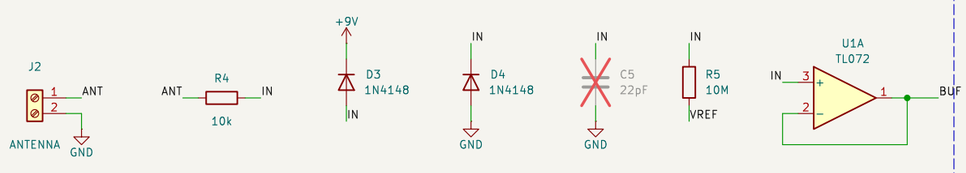
Mounting holes (M3)

4 corner holes for the CAD-week enclosure



3. Antenna input + buffer

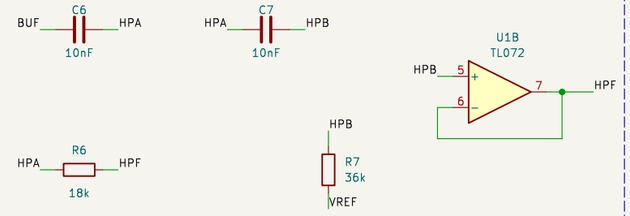
10 M input resistance so the tiny antenna signal isn't loaded down



C5 is DNP (not fitted): only solder it if you hear AM radio.

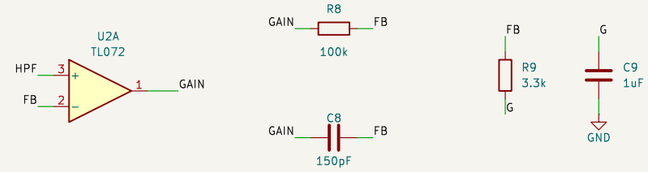
4. Hum filter: 2nd-order high-pass, ~625 Hz

Removes 60 Hz power-line hum



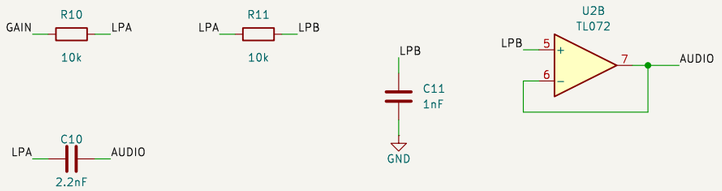
5. Gain stage: x31

G = 1 + R8/R9 = 1 + 100k/3.3k



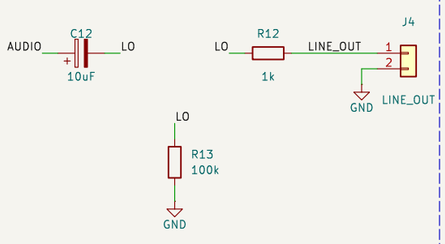
6. High-freq filter: 2nd-order low-pass, ~10.7 kHz

Removes hiss, AM radio and Navy VLF stations



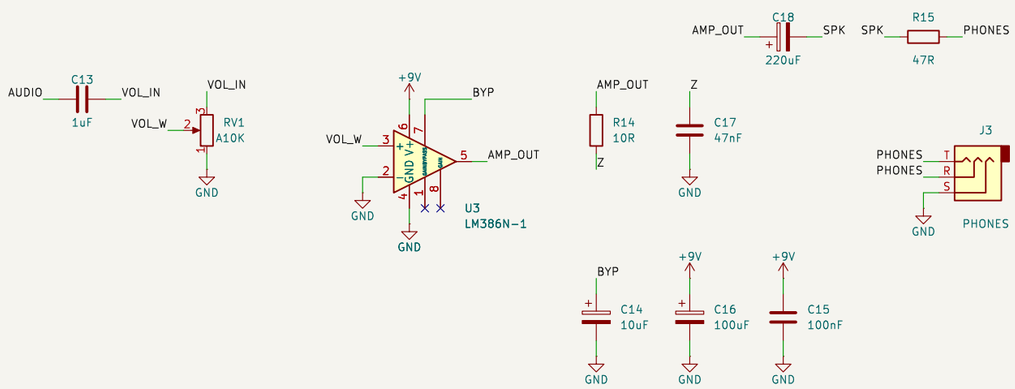
7a. Line out

To the Displays / Synth projects later



7b. Headphone amp (LM386, x20)

Turn the volume to zero before putting headphones on



C15/C16 sit right next to U3 pin 6.

How to read this sheet: every net label with the same name is connected (e.g. all VREF labels are one wire).
Generated as a starting point from the SkyEar Build Guide, Section 5. Run Inspect > Electrical Rules Checker after any edit.

Antenna > buffer > hum filter > gain > low-pass > phones / line out
Week 1: PCB design. 9 V battery powered only.
Hack Club Half Life 2026

Sheet: /
File: skyear.kicad_sch

Title: SkyEar VLF Lightning Receiver

Size: A3	Date: 2026-10-08	**Rev: v1.0**
KiCad E.D.A. 8.0.9		Id: 1/1